Affine Richardson Geometry: Clusters, Links, and Positroid Patches

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1 Introduction

Positroid varieties connect rank conditions on matrices with cluster algebras and knot theory. An *open positroid variety* is a stratum in the Grassmannian $\mathrm{Gr}(k,n)$ of k-planes in $\mathbb{C}^n$, specified by ranks of cyclic column intervals in representative $k \times n$ matrices. Postnikov's work on total positivity organized these strata through planar networks [Pos06]. Knutson–Lam–Speyer related their geometry to *open Richardson varieties*, intersections of opposite Schubert cells in flag varieties [KLS13].

Subsequent work gave these varieties, and more general braid varieties, cluster structures: families of coordinates related by explicit exchange relations [GL23, GLSBS, CGGLSS]. Galashin–Lam connected Richardson geometry to invariants of associated links, obtaining point counts and, in important families, cohomology formulas involving rational q,t-Catalan polynomials [GL24]. These developments make diagrams a means of constructing regular functions and computing topology.

I propose to extend this correspondence to *affine braid Richardson varieties*, defined using periodic flags. They include positroid patches [Sni10]. Bao–Ye's upper-cluster structures provide a starting point [BY25], but identifying our diagrammatic seeds and recovering geometry from periodic links remain open. Exchange relations across a period boundary and winding in the links pose new challenges. Resolving them would extend the coordinate and cohomological methods of finite type to affine geometry.

Problem 1. Develop a periodic diagram model for affine braid Richardson varieties that realizes their coordinate rings as cluster algebras and relates their point counts and cohomology to invariants of associated links.

Building on my affine Deograms and joint results on torus varieties, I will identify cluster seeds, develop affine link invariants, and explain weak separation through explicit seed functions.

2 Past accomplishments

Write $X_{f,\beta}$ for chains of *affine flags*, periodic lattice chains in $\mathbb{C}((z))^n$, with successive relative positions prescribed by a positive affine braid word β and endpoint in the opposite cell indexed by f. An affine permutation is a bijection $f : \mathbb{Z} \to \mathbb{Z}$ with $f(i+n) = f(i) + n$. Reduced β representing g gives $X_{f,\beta} = \mathring{R}_{f,g}$.

My *affine Deograms*, periodic crossing-and-elbow strand diagrams [Mar26], describe the *Deodhar decomposition* into strata $(\mathbb{C}^*)^{e(D)} \times \mathbb{C}^{b(D)}$ [BD94], yielding

$$R_{f,\beta}(q) := \#X_{f,\beta}(\mathbb{F}_q) = \sum_D (q-1)^{e(D)} q^{b(D)}.$$

Local moves give point-count recursions for comparison with link invariants. My bijections connect maximal Deograms, those with the fewest elbows, to rational Dyck paths, lattice paths above a rectangle's diagonal [Mar26].

With Matthew J. Tyler, I characterized *algebraic tori* $(\mathbb{C}^*)^d$ among affine Richardson varieties, extending [GKSB26]. Bruhat order compares reduced subwords; a 2-crown is a subinterval isomorphic to this order on S_3, and ℓ denotes minimum word length.

Theorem 2 (Martinez–Tyler [MT26, Theorem 1.1])**.** *Let $f \leq g$ be affine permutations and $d = \ell(g) - \ell(f)$. The following are equivalent:*

(i) $\mathring{R}_{f,g} \cong (\mathbb{C}^*)^d$*;*

(ii) $[f, g]$ *contains no* 2*-crown;*

(iii) $\mathring{R}_{f,g}$ *has a single Deodhar stratum.*

For a finite positive word β, its *Richardson link* is $L_{u,\beta} = \beta\widehat{\beta(u)^{-1}}$, where $\beta(u)$ is the positive braid lift of u and the hat joins corresponding braid endpoints. A link is *smoothly slice* if its components bound disjoint smoothly embedded disks in the 4-ball. The standard torus rescales coordinate axes. The *Demazure product* $\delta(\beta)$ multiplies the letters of β, omitting steps that decrease ℓ.

Theorem 3 (Martinez–Tyler [MT26, Theorem 1.3])**.** *For a finite positive braid word β and $u \leq \delta(\beta)$, $X_{u,\beta}$ has a dense standard-torus orbit if and only if its Richardson link $L_{u,\beta}$ is smoothly slice.*

These are starting cases for affine coordinates and topology; being an algebraic torus is distinct from having a dense standard-torus orbit [MT26].

3 Research objectives, methods, and significance

Objective 1: Affine cluster structures

I seek generators and exchange relations for the ring $\mathbb{C}[X_{f,\beta}]$ of regular functions. A *cluster seed* consists of algebraically independent rational functions and a *quiver*, a directed graph encoding exchanges. *Mutation* replaces one mutable function and updates the quiver; frozen functions stay fixed. The cluster algebra $\mathcal{A}$ is generated by functions reached by mutation, with frozen inverses. The upper cluster algebra $\mathcal{U}$ consists of functions Laurent in every seed.

Bao–Ye prove $\mathbb{C}[X_{f,\beta}] = \mathcal{U}(\Sigma^{\mathrm{BY}}_{f,\beta})$ [BY25, Theorem 6.5]. Proving the further equality $\mathcal{A} = \mathcal{U}$ would show that cluster variables generate every regular function. With Moriah Elkin and Tyler, I construct candidate quivers $Q^{\mathrm{cand}}_{f,\beta}$ from periodic strand networks with black and white junctions [EMT26].

Theorem 4 (Elkin–Martinez–Tyler [EMT26])**.** *For every positive affine word β and $f \leq \delta(\beta)$, $Q^{\mathrm{cand}}_{f,\beta}$ is locally acyclic.*

Local acyclicity, a finite cover of $\operatorname{Spec}\mathcal{A}$ by cluster localizations with acyclic mutable quivers, implies $\mathcal{A} = \mathcal{U}$ [Mul13]. It remains to identify our candidate with a seed for Bao–Ye's upper-cluster structure; this step is conjectural.

Conjecture 5 (Elkin–Martinez–Tyler [EMT26])**.** *Let β be a positive affine braid word and $f \leq \delta(\beta)$. There is a seed $\Sigma_{f,\beta}$ in $\mathbb{C}(X_{f,\beta})$ with quiver $Q^{\mathrm{cand}}_{f,\beta}$ and its specified frozen vertices such that, with frozen variables inverted,*

$$\mathbb{C}[X_{f,\beta}] = \mathcal{A}(\Sigma_{f,\beta}) = \mathcal{U}(\Sigma_{f,\beta}).$$

I will compare our candidate seed with Bao–Ye's generalized minors, matrix coefficients of fundamental representations. Theorem 2 supplies Laurent polynomial rings for testing *fully frozen* seeds, with no mutable variables. One-mutable-variable examples isolate exchange relations at the affine generator; translation endpoints connect to Objective 3. If the seeds differ, I will compare them through mutations and frozen-monomial rescaling. This would turn periodic diagrams into constructions of regular functions throughout the affine family.

Objective 2: Affine Richardson links

In finite type, links provide effective tools for computing geometric invariants. The *HOMFLY polynomial* $P_L(a, z)$, defined by $P_{\bigcirc} = 1$ and $aP_+ - a^{-1}P_- = zP_0$, recovers Richardson point counts through its normalized highest-a coefficient [GL24, Theorem 2.1]. *Khovanov–Rozansky homology* is a triply graded link invariant whose Euler characteristic recovers this polynomial. Galashin–Lam identify its Hochschild-degree-zero part with torus-equivariant compactly supported Richardson cohomology, after regrading [GL24, Theorem 1.17]. Torus-knot calculations yield mixed Hodge and Betti-number formulas for top-dimensional positroid varieties in $\mathrm{Gr}(k, n)$ with $\gcd(k, n) = 1$, in terms of rational q, t-Catalan polynomials; cluster-geometric Lefschetz isomorphisms then imply Catalan symmetry and unimodality [GL24, Theorem 1.1, Corollary 1.8]. I will seek an affine counterpart of this route from link invariants to cohomology.

My candidate affine link $L_{f,\beta}$ has full periodic lift $\beta\beta(f)^{-1}$ to the universal cover of its ambient space; its evaluation must respect winding.

Problem 6. Make this periodic link construction invariant under diagram moves and construct a HOMFLY evaluation extending the finite-type normalization, with normalized top coefficient

$$\mathcal{H}^{\mathrm{top}}_{f,\beta}(q) = R_{f,\beta}(q) = \#X_{f,\beta}(\mathbb{F}_q).$$

I have checked the identity for all reduced braid lifts of permutations in S_6 using SageMath [Sage]. I will match crossing relations with Deogram recursions, drawing on affine Hecke traces and torus-link models [GN22, GLMon23]. If a polynomial loses winding information, I will retain values in a space of link diagrams modulo crossing relations.

Problem 7. Find an affine counterpart of Theorem 3: does a dense orbit for the diagonal torus action on $X_{f,\beta}$ admit a characterization by smooth surfaces bounded by $L_{f,\beta}$?

I will specify the ambient four-manifold and surface conditions compatible with winding. Our finite proof compares torus weights with a surface assembled from the same braid crossings [MT26]. Periodic versions, first for short words and torus examples, may extend this topological characterization of a geometric action.

A homological refinement should recover the point-count identity by graded Euler characteristic. Comparisons with equivariant cohomology for tori and short words will test grading shifts and torus factors, toward affine cohomology formulas and geometric interpretations of Deogram and Dyck-path statistics.

Objective 3: Affine positroid combinatorics

Le-diagrams, dot fillings of Young diagrams with a southeast-completion rule for southwest–northeast pairs, index cells in the totally nonnegative Grassmannian, where all maximal minors

are nonnegative, and give positive coordinates [Pos06]. I will extend this theory to affine geometry and determine when Plücker coordinates describe the seed functions.

The homogeneous *Plücker coordinate* Δ_I is the maximal minor in columns I. For an open positroid variety Π_f°, Snider identifies the patch $U_{f,I} = \Pi_f^\circ \cap \{\Delta_I \neq 0\}$ with $\mathring{R}_{f,t_I}$, where t_I shifts the coordinates in I by one period [Sni10]. A *plabic graph* is a planar graph in a disk with black and white internal vertices and k-subset face labels. Freezing a face labeled I gives the cluster model for this localization.

Two k-subsets I, J are *weakly separated* if no chord joining elements of $I \setminus J$ crosses a chord joining elements of $J \setminus I$ in the cyclic order on $[n]$. The *Grassmann necklace* $\mathcal{I}_f$ consists of the positroid's lexicographically minimal bases in each cyclic order. A nonzero index I occurs as a plabic face label exactly when it is weakly separated from every $J \in \mathcal{I}_f$ [OPS15, Theorem 1.5].

With Elkin and Tyler, I extend this model using periodic diagrams and plabic graphs in a punctured disk [EMT26].

Theorem 8 (Elkin–Martinez–Tyler [EMT26])**.** *Let $\mathcal{R}$ be the region associated with a periodic $k \times (n-k)$ path. A periodic dot filling of $\mathcal{R}$ is an affine Le-diagram if and only if every southwest–northeast pair of dots whose square is contained in $\mathcal{R}$ has a dot at its southeast corner.*

The *additional Le-constraint* forbids pairs whose square leaves $\mathcal{R}$. *Soap films* are surfaces traced along strands from edges joining them in a 3D plabic graph [GLSBS]; our periodic analogues index the candidate seed functions.

Theorem 9 (Elkin–Martinez–Tyler [EMT26])**.** *Let $D_{f,I}$ be the affine Deogram of a nonempty patch $U_{f,I}$. The following are equivalent:*

(i) *$D_{f,I}$ satisfies the additional Le-constraint;*

(ii) *I is weakly separated from every $J \in \mathcal{I}_f$;*

(iii) *the soap films of $D_{f,I}$ do not intersect.*

This explains weak separation through nonintersection and suggests Plücker seed coordinates.

Conjecture 10 (with Elkin and Tyler [EMT26])**.** *Suppose $D_{f,I}$ satisfies the additional Le-constraint, and fix $J_0 \in \mathcal{I}_f$. For each variable x_v of its candidate affine seed, there are $J_v \in \binom{[n]}{k}$ and an invertible frozen monomial m_v such that*

$$x_v = m_v \frac{\Delta_{J_v}}{\Delta_{J_0}}.$$

I will compare these functions with freezing the face labeled I, using face-preserving moves [OS17]. When weak separation fails, no such face exists, although normalized Δ_I is a cluster monomial, a product of variables from one cluster with frozen factors invertible [MMSV24, Theorem 8.9]. Starting with one failed additional constraint, I will compute the functions represented by intersecting films and compare them with localization at those factors. This should explain the Plücker description for nonintersecting films and the functions needed beyond it.

4 Career development

The Research Instructorship will develop my independent program at the intersection of combinatorics, geometry, and topology. Regular meetings with Sherman-Bennett will build the representation-theoretic expertise needed to prove coordinate-ring results; learning link homology and surface methods at Davis will broaden the geometric questions I can pursue.

Organizing a learning seminar and presenting in the Algebra and Discrete Mathematics seminar will strengthen my exposition and develop collaborations. Teaching will build course-design and assessment skills, while undergraduate and beginning graduate mentorship will prepare me to formulate accessible projects and support independent research.

5 Sponsoring scientist and host institution

Melissa Sherman-Bennett's work on 3D plabic graphs [GLSBS], comparisons of cluster constructions [Comp25], and Richardson tori [GKSB26] makes her an ideal sponsor for the transition from diagrams to coordinate-ring statements. Her expertise will guide the seed comparisons and interpretation of affine torus cases.

Eugene Gorsky's work on braid varieties, link homology, and affine Hecke traces [CGGLSS, GN22], together with Roger Casals's work on cluster structures and contact and symplectic geometry [CGGLSS, Comp25], makes UC Davis especially well suited to the project's geometric and topological questions.

6 Broader Impacts

Undergraduate research shaped my decision to pursue graduate study. Since 2023, I have mentored eight undergraduates through the UCLA Directed Reading Program and served on its organizing committee. In spring 2026, Olha Shevchenko and I co-mentored three of these students on a project that led to a joint preprint on unimodality of q-Fibonomial coefficients [CIMSY26].

At Davis, I aim to mentor one or two undergraduates annually through the Davis Math Lab and directed reading. Deograms and rational Dyck paths offer accessible enumeration and bijection problems. I will tailor projects to students' preparation, guiding them from examples to proofs and presentation. I will also mentor graduate students through joint reading and research, as well as navigating through academia in mathematics.